Gurvir Singh

Third-Year B.Tech Student • Seeking an AI Engineering Internship

Computer Science & Engineering
(Artificial Intelligence & Data Science)
GNA University, Punjab, India

gurvir.singh.panesar@gmail.com • +91 9592592881
github.com/GurvirSinghH
gurvirsingh.in

Profile

Computer Science (AI & Data Science) student who builds working AI tools end to end in Python: an LLM assistant that writes code, runs it and feeds errors back to the model to fix; an ML app that flags anomalies in system logs; and a crawler that turns a website into a retrieval-ready dataset.

Projects

Neural Forge (AI Blender Assistant) (v1.0.0) *Python, Blender Python API, LLM APIs*

- Blender add-on that turns natural-language prompts into Blender Python scripts, with one client for Gemini, OpenAI, local Ollama models and any OpenAI-compatible endpoint.
- Adds scene context (objects, selection) to the prompt, runs the generated code and sends errors back to the model to fix them: an agent-style generate, run and repair loop.

AI Log Intelligence Platform (in development) *Python, Streamlit, Pandas, Scikit-learn, Plotly*

- Streamlit app that detects and parses Linux auth/syslog and Apache access/error logs, with summary statistics, keyword search, Plotly charts and an optional Gemini-drafted incident report.
- Label-encoded process, module and host fields plus PID, message length and hour feed an Isolation Forest anomaly detector; K-Means then clusters the flagged entries.

College Knowledge Base Crawler *Python, Requests, BeautifulSoup, lxml, SQLite*

- Multi-threaded crawler (worker pool and task queue) with robots.txt compliance, a shared rate limit, URL normalization, and retries with exponential backoff that respect Retry-After.
- Extracts page content and metadata to JSON, downloads linked documents and records crawl state in SQLite, as the data layer for a RAG pipeline (retrieval step not built yet).

Home Energy Monitor (in progress) *ESP32, PZEM-004T, Fedora Linux*

- An ESP32 with a PZEM-004T meter sends household power readings to my self-hosted Fedora server (message broker and logger); a simulated dry run works. Next: recognise which appliance is running from its power signature and estimate the monthly bill.

Also built: CodeEditor (Java Swing editor, MySQL over JDBC), CSV Cleaner (Python data cleaning), ML Lab (Scikit-learn notebooks).

Technical Skills

Languages	Python (primary), C++, C, Java, HTML, CSS
AI and LLMs	Gemini, OpenAI, Ollama and OpenAI-compatible APIs; prompt and context design; LLM code generation with error feedback
ML and data	Scikit-learn (Isolation Forest, K-Means, decision trees, regression), NumPy, Pandas, Matplotlib, Seaborn, Plotly
Tools	Streamlit, Git/GitHub, Jupyter, SQLite, MySQL, Linux (Fedora), SSH, Tailscale
Fundamentals	Data structures and algorithms (practised in Python and C++)

Education

GNA University **Expected 2028**

B.Tech, Computer Science & Engineering (AI & Data Science), third year CGPA: 8.13/10

Coursework: Data Visualization, Software Engineering, Theory of Computation, Computer Graphics

Currently Learning

- **AI engineering:** working through AI Engineering from Scratch, an open-source course from math and ML foundations through deep learning to LLMs, RAG and agents.
- **Physics-informed ML:** building known physics into model training, e.g. via the loss function.